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IMPURITY TRANSPORT AND RADIATION AT THE STELLARATOR WENDELSTEIN 7-X

Promotionskolloquium

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Foto: Dr. Matthias Otte

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CONTENTS

MOTIVATION

NUCLEAR FUSION: ENERGY CHALLENGE

- Lawson criterion for net energy gain:

$$n_e T \tau_E \geq \frac{12 f_{\mathrm{tot}}}{\langle \sigma_{\mathrm{DT}} v \rangle f_H^2 E_\alpha - 4 L_Z(T)} T^2$$

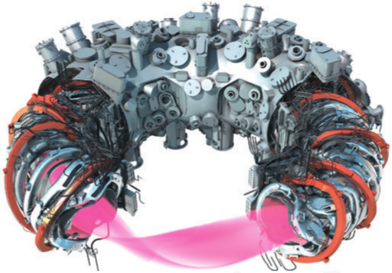
- Requires: high temperature ($T \sim 10\,\mathrm{keV}$ to $20\,\mathrm{keV}$), density, and confinement time

Triple Product

Key figure of merit for fusion performance combining density, temperature, and energy confinement time

WENDELSTEIN 7-X

- World's largest stellarator (major radius $5.5\,\mathrm{m}$, minor radius $0.53\,\mathrm{m}$)
- 70 superconducting magnets: 50 non-planar & 20 planar coils
- Volume $30\,\mathrm{m}^3$, $B_{\mathrm{max}} = 3\,\mathrm{T}$
- Island divertor configuration (5/6, 5/5, 5/4)
- Optimized for reduced neoclassical transport
- Heating power up to $14\,\mathrm{MW}$ ECRH



Key Challenge

Managing heat loads on plasma-facing components during long, high performance plasma

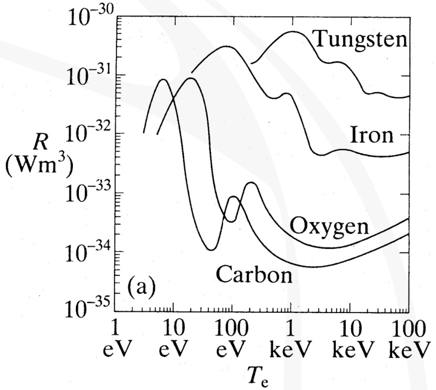
PLASMA RADIATION AND TRANSPORT

Transport Mechanisms:

- Classical: collisional diffusion
- Neoclassical: trapped particles, $\propto 1/\nu$
- Anomalous: turbulence (dominant)

Radiation:

- Bremsstrahlung: $P_{\text{Brems}} \propto n_e^2 T^{1/2}$
- Line radiation: atomic transitions
- Impurity effects on plasma performance



Impurity Seeding

Injection of low-Z (He, N_2) or high-Z (Ne, Ar) impurities to enhance edge radiation and cooling

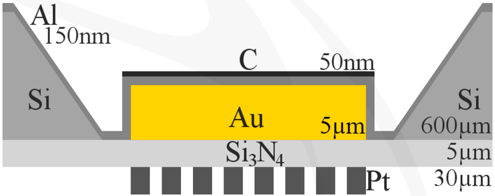
BOLOMETER DIAGNOSTIC

EUROfusion IPP

BOLOMETRY AT W7-X

Metal Resistor Bolometers:

- Thin metal (gold) film absorbers
- Wheatstone bridge circuit configuration
- Measures total radiation power (broadband)
- Multicamera system: HBC & VBCI/r (32 + 2×24 channels)



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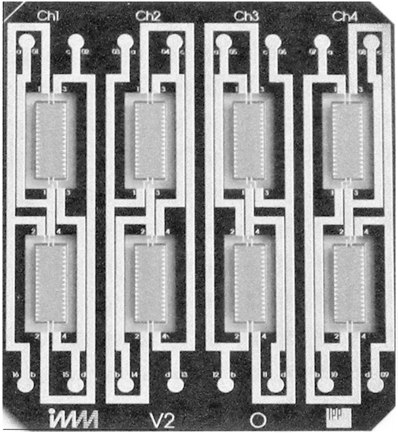

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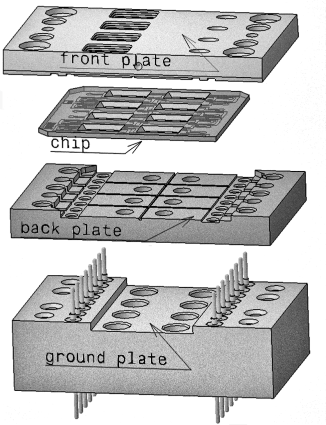

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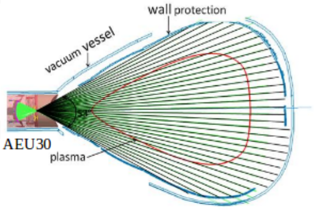
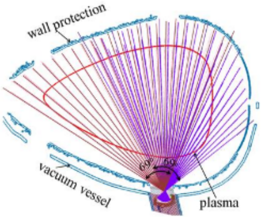


horizotal bolomter camera (HBC)



vertical bolometer camera (VBC)

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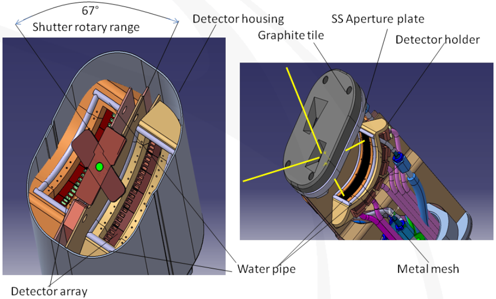
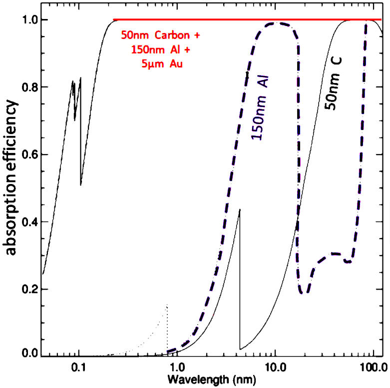


PERFORMANCE

Performance Characteristics:

- In-situ ohmic heating calibration (F_{M}, f_{M})
- Temporal resolution: $\{0.4, 0.8, 1.6, ..., 6.4\}$ ms
- Measurement uncertainty (*Gauss*): $1.6\,\mu$W $\leftrightarrow$ SNR > 1000
- Spatial resolution: $\sim$5 cm at magnetic axis
- Near unity absorption efficiency: 600-0.2 nm



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Power Per Detector:

$$P_M = F_M \cdot \left(\frac{d(\Delta \widetilde{U}_M)}{dt} + f_M \Delta \widetilde{U}_M \right)$$

Global Radiation Power:

$$P_{\mathrm{rad}} = \frac{V_{\mathrm{P,tor}}}{V_C} \sum_M \frac{P_M V_M}{K_M}$$

PERFORMANCE

Chord Brightness

Line-integrated measurement along each detector's line of sight provides radial profile information ($P_{\text{chord}} \sim P_{\text{M}}/K_{\text{M}}$)

Applications:

- Power balance calculations
- **Tomographic reconstruction**
- **Real-time feedback control**

$\rightarrow$ controlled radiative cooling in the plasma edge

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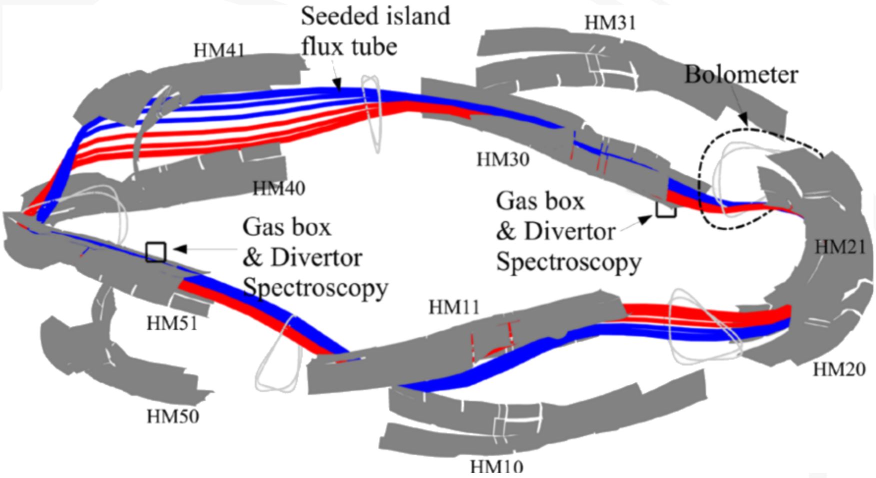
REAL-TIME RADIATION FEED-BACK CONTROL

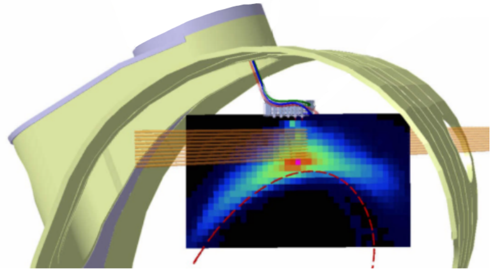
FEEDBACK SYSTEM



FEEDBACK SYSTEM

Architecture:

- Integrated data acquisition hardware & optimized control software
- Twin proxies and dedicated signal outputs
- Thermal gas valve actuation with multipurpose/-diagnostic PID controller



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Radiation Prediction Proxies:

A: *Channel selection S*

$$P_{\text{pred}}^{(1)} = \frac{V_{\text{P,tor}}}{V_S} \sum_{\text{M}}^{S} \frac{P_M V_M}{K_M}$$

(e.g. $S_{\text{VBC}} = \{61, 86, 78, 73, 70, 65\}$)

B: *Single channel*

$$P_{\text{pred}}^{(2)} \propto \Delta \widetilde{U_M}$$

(dimensionless, minimal overhead)

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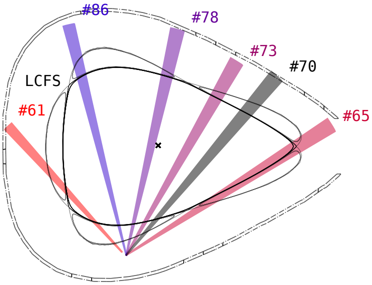
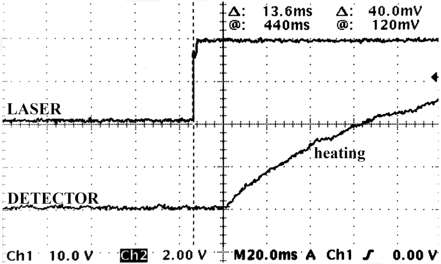


Feedback System

System Performance:

- FIFO smoothing in $P^{(1)}_{\mathrm{pred}}$: $\sim$M · Δt/2 (M$=10$, Δt$=1.6$ ms)
- Minimum acquisition latency 13.6 ms
- Total system latency: 150 ms to 400 ms (dominated by plasma response)

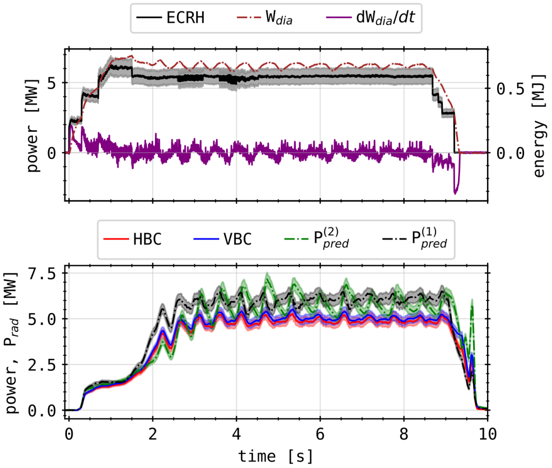


(latency measurement with laser diode)

EXPERIMENTAL ACHIEVEMENTS

XP20181010.32

- Duration $9.2\,\mathrm{s}$
  $P_{\mathrm{ECRH}}^{\mathrm{max}} = 6.23\,\mathrm{MW}$ (O2-mode)
- Total input energy $55\,\mathrm{MJ}$
- Radiation fraction PID setpoint $f_{\mathrm{rad}} \sim 90\%$ off $P_{\mathrm{pred}}^{(1)}$
- 5-channel VBC subset $S = \{65, 78, 61, 73, 86\}$



EUROfusion IPP

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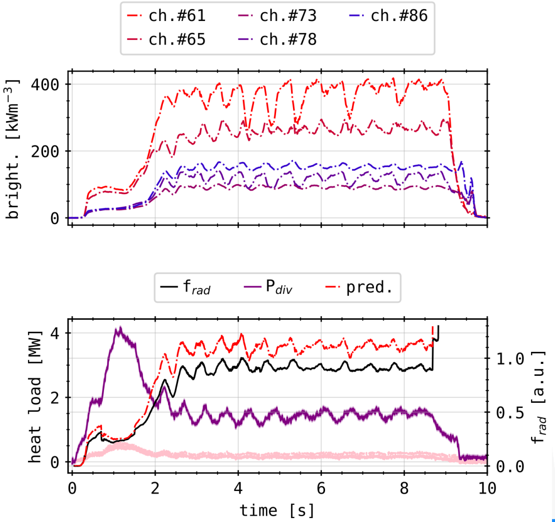

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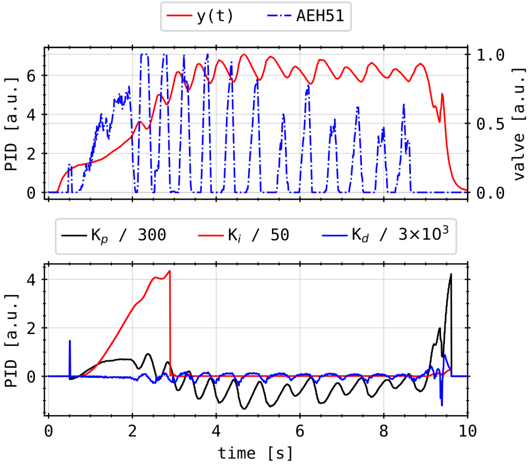

EUROfusion
IPP

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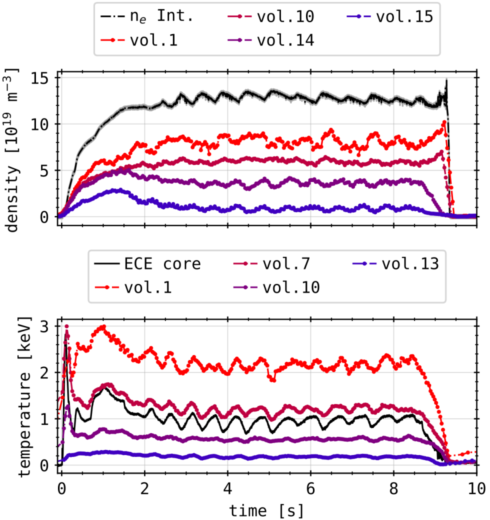
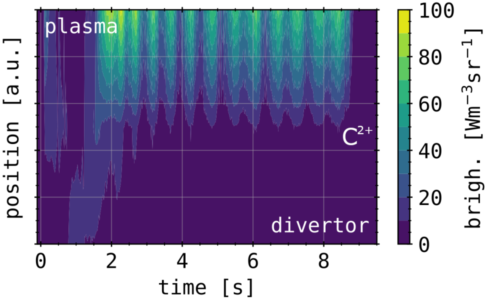


EXPERIMENTAL ACHIEVEMENTS

XP20181010.32

- Peak $f_{\mathrm{rad}} \gtrsim 100\%$, cycling with gas inlet
- Target heat load reduction factor >2
- C^{2+} detachment visible at $f_{\mathrm{rad}} \sim 50\%$



Key Result

Demonstrated stable, real-time feedback-controlled radiative cooling with $f_{\mathrm{rad}} \geq 85\%$ without terminal plasma disruption and significant target heat load reduction

LINE-OF-SIGHT SENSITIVITY

CHANNEL SELECTION SENSITIVITY

- Evaluate LOS/channel efficacy for feedback (generalised local radiation sensitivity)
- Data set of all feedback discharges from OP1.2b
- Applied 5 different metric (linear & weighted deviation, FFT, ...)

Weighted deviation:

$$\varphi(t) = 1 - \frac{\|P_{\mathrm{pred}} - P_{\mathrm{rad}}\|(t)}{P_{\mathrm{rad}}(t)}$$

$$\vartheta = \frac{1}{T}\int_T \varphi(t)\,\mathrm{d}t$$

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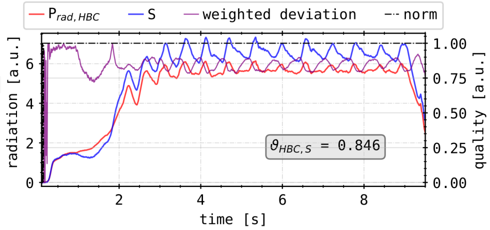
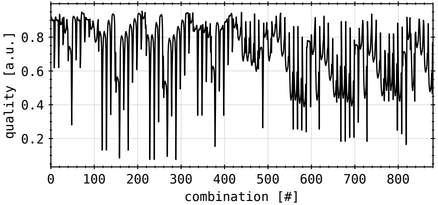
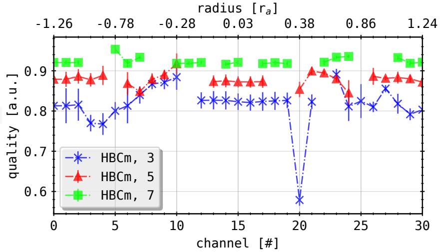


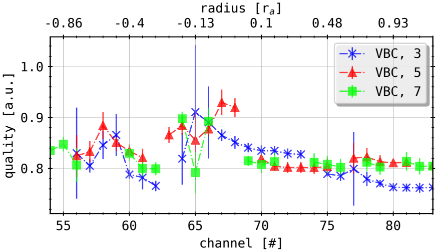


CHANNEL SELECTION SENSITIVITY

Optimal Channel Selection:

- No single "best" set exists
- Robust selection achieves ≥85% prediction accuracy
- VBC channels generally outperform HBC for edge sensitivity
- 3-5 channels sufficient for reliable P_{pred}





Takeaway

Detectors viewing close to separatrix and SOL most viable for feedback purposes

TOMOGRAPHY

MINIMUM FISHER REGULARIZATION

Measurement:

$$\mathbf{T}\,\vec{x} = \vec{b}$$

Bolometer tomography:

- 61 line-integrated measurements $\vec{b}$
- 2D emissivity $\vec{x}$: $\sim$4500 pixels
- *Geometry* $\mathbf{T}$ has high condition
- ill-posed, requires regularization

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Problem:

$$\min_{\eta} \left\| \left(\mathbf{T}^{\intercal}\mathbf{T} + \mu^{-1}\mathbf{H}\right)^{-1} \mathbf{T}^{\intercal}\vec{b} \right\| \to \vec{x}$$

Fisher Information:

$$I_{\mathrm{F}} = \int \frac{1}{x}\left(\frac{\partial x}{\partial \vec{r}}\right)^2 d\vec{r} \leftrightarrow \mathbf{H} \sim \frac{1}{x}\boldsymbol{\Delta}^{\intercal}\boldsymbol{\Delta}$$

limiting error on reconstruction:

$$\sigma_{\mathrm{x}} \leq \frac{1}{I_{\mathrm{F}}}$$

MINIMUM FISHER REGULARIZATION

Radially Dependent Anisotropy:

- Tailored weighting to poloidal gradient
- $k_{\mathrm{ani}} < 1$: favours anisotropy
- $k_{\mathrm{ani}} > 1$: favours smooth emissivity

$$\mathbf{H} \rightarrow \mathbf{H}_{\mathrm{ani}}$$

$$\boldsymbol{\Delta} = \boldsymbol{\nabla}_{\mathrm{r}}^{\mathsf{T}}(\cdot)\boldsymbol{\nabla}_{\mathrm{r}} + \boldsymbol{\nabla}_{\vartheta}^{\mathsf{T}}(\cdot)\boldsymbol{\nabla}_{\vartheta}$$

$$\boldsymbol{\nabla}_{\vartheta} \rightarrow \tilde{\boldsymbol{\nabla}}_{\vartheta} = \frac{1}{k_{\mathrm{ani}}\Delta r}\mathbf{D}_{\mathrm{dia}}^{(0)}\boldsymbol{\nabla}_{\vartheta}$$

MFR + RDA Advantages

- Robust to noisy data
- Incorporates a priori knowledge
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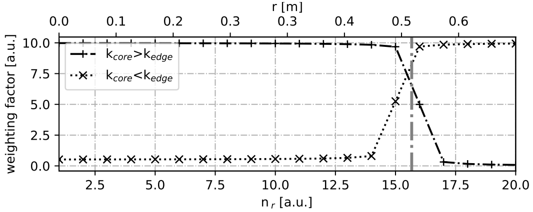

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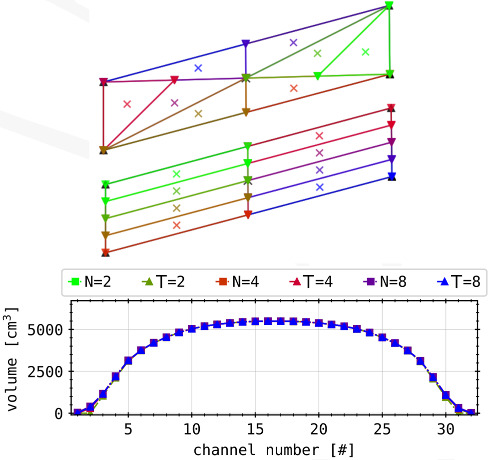
BENCHMARKING



Geometry Perturbation Test:

- Previous and future calculations based on as-designed geometry

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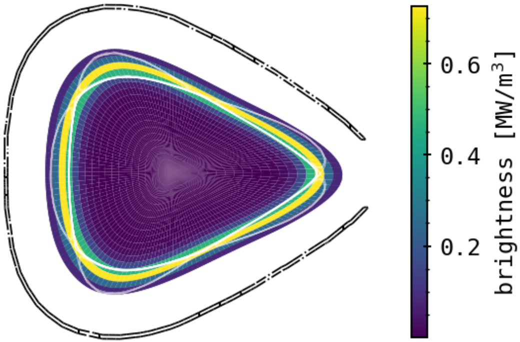

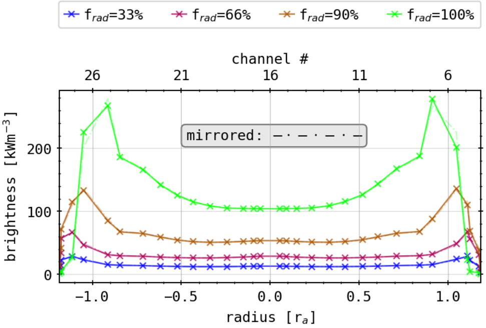

$(f_{\mathrm{rad}} \sim 100\%)$

BENCHMARKING

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- Tests for symmetrical chord profile yield artificial camera geometry

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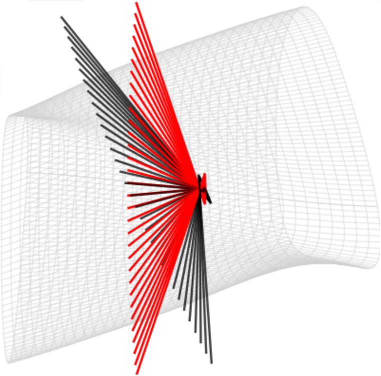
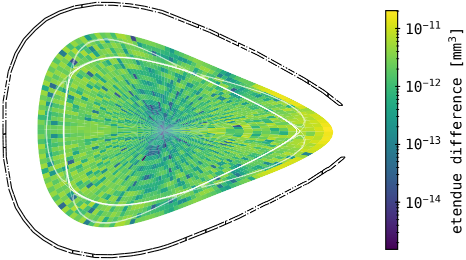
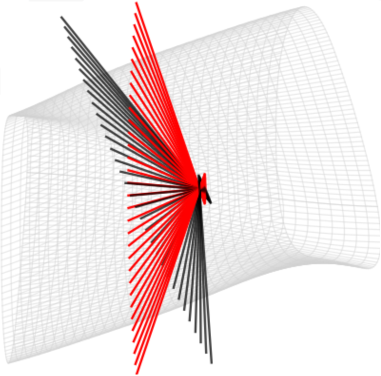
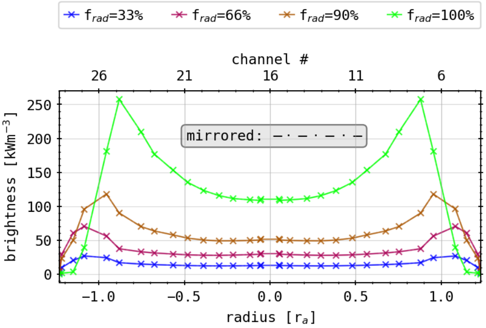




$$\|\mathbf{T} - \mathbf{T}_{\mathrm{sym}}\|$$

BENCHMARKING





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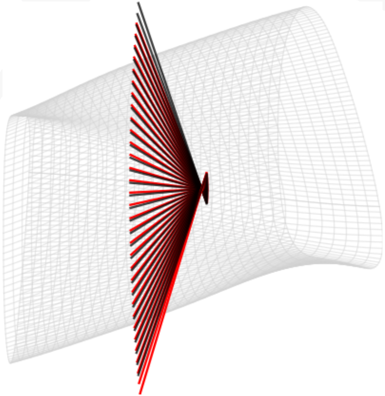

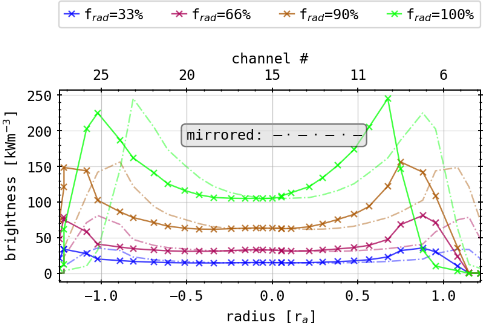

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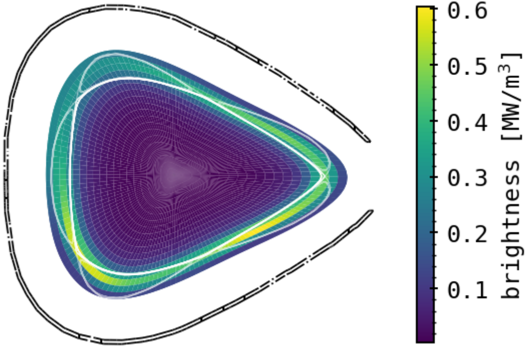
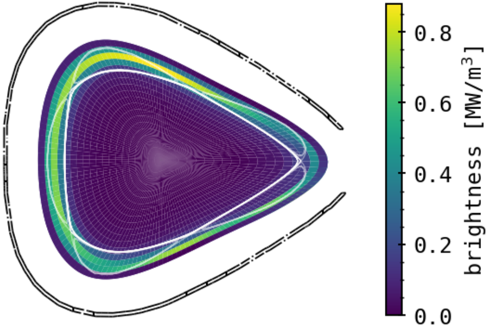


using $\mathbf{T}_{\mathrm{sym}}$



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Results

- Discretization of system is computationally stable
- Intrinsic asymmetry of same order of magnitude as potential, unknown geometric pertubation - non-negligible
- Worst case machine warp scenario $\sim 0.2 r_{\mathrm{a}}$ shift

PHANTOM RECONSTRUCTIONS

Purpose

Regularized, parametric inversion methods need (experienced) tailoring towards their respective application - if not characteristic distributions.

PHANTOM RECONSTRUCTIONS

Benchmarks:

- 12+ synthetic emissivities: Gaussian, rings, anisotropies, convolutions...
- k_{ani} & phantom variations studies (radius, intensity etc.)
- Experimentally motivated distributions
- Performed for only one magnetic grid configuration & LOS geometry

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Metrics:

$$\chi^2 = \frac{1}{N_{\mathrm{ch}}} \left\| \left(\vec{b}_{\mathrm{phan}} - \vec{b}_{\mathrm{tom}} \right) \vec{\sigma}_{\mathrm{bol}}^{-1} \right\|$$

$$\rho_{\mathrm{c}} = \frac{1}{\sigma_{pha}\sigma_{tom}} \mathrm{cov}\left(\vec{x}_{\mathrm{phan}}, \vec{x}_{\mathrm{tom}}\right)$$

$$M_{\mathrm{SD}} \sim \left\| \frac{\vec{x}_{\mathrm{tom}} - \vec{x}_{\mathrm{phan}}}{\left\| x_{\mathrm{phan}} \right\|} \cdot \vec{A}^{\mathsf{T}} \right\|$$

$$P_{\mathrm{rad,X}}^{\mathrm{2D}} = \vec{x}_{\mathrm{X}}^{\mathsf{T}} \cdot \vec{A}, \quad X = \{\mathrm{phan},\ \mathrm{tom}\}$$

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- Performed for only one magnetic grid configuration & LOS geometry

Metrics:

$$\boxed{\chi^2 = \frac{1}{N_{\text{ch}}} \left\| \left(\vec{b}_{\text{phan}} - \vec{b}_{\text{tom}} \right) \vec{\sigma}_{\text{bol}}^{-1} \right\|}$$

$$\rho_c = \frac{1}{\sigma_{pha}\sigma_{tom}} \text{cov} \left(\vec{b}_{\text{phan}}, \vec{b}_{\text{tom}} \right)$$

$$M_{\text{SD}} \sim \left\| \frac{\vec{x}_{\text{tom}} - \vec{x}_{\text{phan}}}{\| x_{\text{phan}} \|} \cdot \vec{A}^{\mathsf{T}} \right\|$$

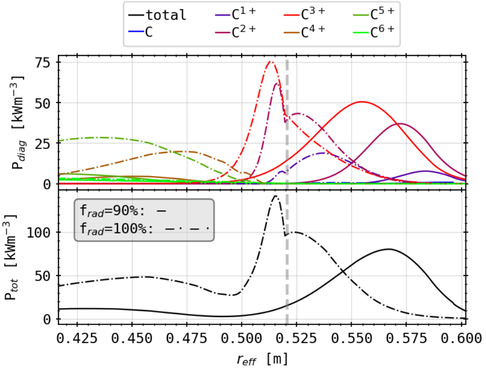
$$P_{\text{rad,X}}^{\text{2D}} = \vec{x}_{\text{X}}^{\mathsf{T}} \cdot \vec{A}, \quad X = \{\text{phan, tom}\}$$

PHANTOMS: EXAMPLE

STRAHL IMPURITY TRANSPORT MODELING

STRAHL Code:

- 1D impurity transport and radiation simulation
- Solves radial continuity for each ionisation stage
- Experimental input data: pertubation study in transport coefficients, profile smoothing, source model etc.
- Valid in confinement, simple ansatz outside LCFS



EUROfusion IPP

STRAHL IMPURITY TRANSPORT MODELING

STRAHL Code:

- 1D impurity transport and radiation simulation
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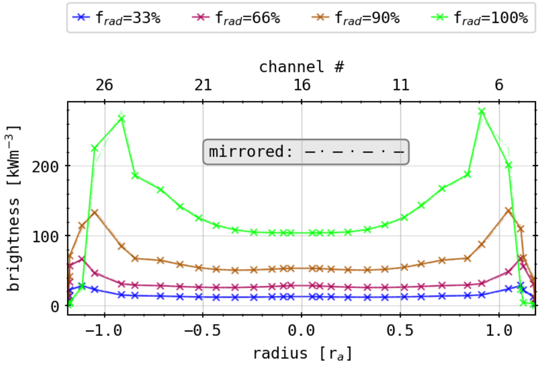
Key Results:

- Carbon dominates emissivity ($C^{2/3+}$, x10^2+ of Oxygen levels)
- Inward shift of radiation for $f_{\mathrm{rad}} \to 1$ toward $r \sim 0.95 r_{\mathrm{a}}$
- Radiation moves from outside to inside separatrix

STRAHL IMPURITY TRANSPORT MODELING



Forward Modeling:

- Chord brightness from STRAHL results
- Compare with experimental measurements
- Validate assumptions about impurity radiation and transport

Discrepancy

Forward calculations show asymmetry - less than experiments - in confinement, suggesting intrinsic geometry effects

STRAHL IMPURITY TRANSPORT MODELING